BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: MRD-XI-21(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-XI-21(A)

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only, Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. .....are responsible for functional diversity of the cell membrane.

(A) Lipids
(B) Proteins
(C) Carbohydrates
(D) All these

Q2. Golgi complex consists of units called.....

(A) Peroxysome
(B) Dictyosomes
(C) Cristae
(D) Microbodies

Q3. Which of the given is an example of Purines?

(A) Adenine
(B) Thymine
(C) Cytosine
(D) Uracil

Q4. Aminoacids mainly differ from each other by the difference in their.....

(A) Aminogroup
(B) Carboxyl group
(C) R.group
(D) Alpha group

Q5. Enzyme of the arctic flea work at.....

(A) −10°
(B) −9°
(C) −5°
(D) −1°

Q6. The optimum pH value for pepsin to work is.....

(A) 4
(B) 5
(C) 4.5
(D) 1.5

Q7. The four rings (Tetra Pyrol Ring) in chlorophyll molecule is known as.....

(A) Granum
(B) Stroma
(C) Tail
(D) Porphyrin

Q8. During Non Cyclic Electron Transport of Light reaction, electrons from PS-II are directly captured by.....

(A) Fructose
(B) Lactose
(C) PQ
(D) RuBP

Q9. The example of spherical virus is.....

(A) TMV
(B) Polio Virus
(C) Bacteriophage
(D) None of these

Q10. Refrigeration at temperature 0°C to.....reduces activities of microbes.

(A) 5
(B) 6
(C) 7
(D) 8

Q11. Yeast belongs to which group of fungi?

(A) Myxomycota
(B) Ascomycota
(C) Oomycota
(D) Water molds

Q12. In bryophytes the male reproductive organ is known as:

(A) Stamen
(B) Carpel
(C) Antheridium
(D) Archegonium

Q13. Word Porifera means.....

(A) holes bearer
(B) spiny skinned
(C) Soft body
(D) Little rings

Q14. Garden snail belongs to..... phylum.

(A) Arthropoda
(B) Chordate
(C) Mollusca
(D) Coelenterata

Q15. Movements of plants organs, in response to moisture is called...

(A) Thigmo-trophism
(B) Hydro-trophism
(C) Chemo-trophism
(D) Photo-trophism

Q16. The final part of large intestine is.....

(A) Ceacum
(B) Colon
(C) Duodenum
(D) Rectum

Q17. Lymphatic vessels in cervical region does not collect lymph from.....region.

(A) Scalp
(B) Face
(C) Intestine
(D) Pharynx

Q18. Surgical removal of thymus of a new born shall results in failure to produce.....

(A) Monocyta
(B) Monocytes
(C) T-lymphocytes
(D) Basophiles

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 40

QII. Write short answers of any ten (10) of the following parts. All parts carry equal marks.

1. Write a short note on Ribosome.
2. Enlist the differences between DNA and RNA.
3. Discuss the affect of temperature on the rate of enzyme action.
4. How Electron Transport Chain occur during aerobic respiration?
5. Describe the events of Lytic cycle of Bacteriophage.
6. Draw a labeled diagram of the structure of bacterial cell, showing all its parts.
7. Write the general characteristics of Fungi.
8. Briefly explain the life cycle of a flowering plant.
9. Write the distinguishing characteristics of phylum Echinodermata.
10. How movements of water through roots occur by different pathways?
11. Write comparison between the structure and function of liver and pancreas.
12. Discuss briefly the structure and function of various parts of human heart.
13. Discuss the inflammatory responses as the non specific defense mechanism.

Section – C

Maximum Marks: 27

> Note: Answer any three (3) questions. All questions carry equal marks.

QIII.

(a) [5] Compare between the structure and role of chloroplast and mitochondria.

(b) [4] Write any four biological importance of water.

QIV.

(a) [5] Define hepatitis, also write a short note on hepatitis B.

(b) [4] Describe the phases of growth in bacteria.

QV.

(a) [4] Write a short note on the mycorhizal association.

(b) [5] Define macronutrients. Make a chart of macronutrients and show its importance.

QVI. Discuss any two of the following

(i) [4.5] Structure of Human Stomach

(ii) [4.5] Role of lymphatic system in human body

(iii) [4.5] Types of immunity